

Conditional Probability and Independent Events

32. The automobile industry sold 657,000 vehicles in the United States during January 2009 (*The Wall Street Journal*, February 4, 2009). This volume was down 37% from January 2008 as economic conditions continued to decline. The Big Three U.S. automakers—General Motors, Ford, and Chrysler—sold 280,500 vehicles, down 48% from January 2008. A summary of sales by automobile manufacturer and type of vehicle sold is shown in the following table. Data are in thousands of vehicles. The non-U.S. manufacturers are led by Toyota, Honda, and Nissan. The category Light Truck includes pickup, minivan, SUV, and crossover models.

Manufacturer	Type of Vehicle	
	Car	Light Truck
	U.S.	Non-U.S.
	87.4	193.1
	228.5	148.0

- Develop a joint probability table for these data and use the table to answer the remaining questions.
 - What are the marginal probabilities? What do they tell you about the probabilities associated with the manufacturer and the type of vehicle sold?
 - If a vehicle was manufactured by one of the U.S. automakers, what is the probability that the vehicle was a car? What is the probability it was a light truck?
 - If a vehicle was not manufactured by one of the U.S. automakers, what is the probability that the vehicle was a car? What is the probability it was a light truck?
 - If the vehicle was a light truck, what is the probability that it was manufactured by one of the U.S. automakers?
 - What does the probability information tell you about sales?
33. In a survey of MBA students, the following data were obtained on “students’ first reason for application to the school in which they matriculated.”

Enrollment Status	Reason for Application			
		School Quality	School Cost or Convenience	Other
	Totals			
Full Time	421	393	76	890
Part Time	400	593	46	1039
Totals	821	986	122	1929

- Develop a joint probability table for these data.
- Use the marginal probabilities of school quality, school cost or convenience, and other to comment on the most important reason for choosing a school.

- c. If a student goes full time, what is the probability that school quality is the first reason for choosing a school?
 - d. If a student goes part time, what is the probability that school quality is the first reason for choosing a school?
 - e. Let A denote the event that a student is full time and let B denote the event that the student lists school quality as the first reason for applying. Are events A and B independent? Justify your answer.
34. The U.S. Department of Transportation reported that during November, 83.4% of Southwest Airlines' flights, 75.1% of US Airways' flights, and 70.1% of JetBlue's flights arrived on time (*USA Today*, January 4, 2007). Assume that this on-time performance is applicable for flights arriving at concourse A of the Rochester International Airport, and that 40% of the arrivals at concourse A are Southwest Airlines flights, 35% are US Airways flights, and 25% are JetBlue flights.
- a. Develop a joint probability table with three rows (airlines) and two columns (on-time arrivals vs. late arrivals).
 - b. An announcement has just been made that Flight 1424 will be arriving at gate 20 in concourse A. What is the most likely airline for this arrival?
 - c. What is the probability that Flight 1424 will arrive on time?
 - d. Suppose that an announcement is made saying that Flight 1424 will be arriving late. What is the most likely airline for this arrival? What is the least likely airline?
35. According to the Ameriprise Financial Money Across Generations study, 9 out of 10 parents with adult children ages 20 to 35 have helped their adult children with some type of financial assistance ranging from college, a car, rent, utilities, credit-card debt, and/or down payments for houses (*Money*, January 2009). The following table with sample data consistent with the study shows the number of times parents have given their adult children financial assistance to buy a car and to pay rent.

		Pay Rent	
		Yes	No
Buy a Car	Yes	56	52
	No	14	78

- a. Develop a joint probability table and use it to answer the remaining questions.
- b. Using the marginal probabilities for buy a car and pay rent, are parents more likely to assist their adult children with buying a car or paying rent? What is your interpretation of the marginal probabilities?
- c. If parents provided financial assistance to buy a car, what is the probability that the parents assisted with paying rent?
- d. If parents did not provide financial assistance to buy a car, what is the probability the parents assisted with paying rent?
- e. Is financial assistance to buy a car independent of financial assistance to pay rent? Use probabilities to justify your answer.
- f. What is the probability that parents provided financial assistance for their adult children by either helping buy a car or pay rent?

36. Jerry Stackhouse of the National Basketball Association's Dallas Mavericks is the best free-throw shooter on the team, making 89% of his shots (ESPN website, July, 2008). Assume that late in a basketball game, Jerry Stackhouse is fouled and is awarded two shots.
- What is the probability that he will make both shots?
 - What is the probability that he will make at least one shot?
 - What is the probability that he will miss both shots?

37. Visa Card USA studied how frequently young consumers, ages 18 to 24, use plastic (debit and credit) cards in making purchases (Associated Press, January 16, 2006). The results of the study provided the following probabilities.

- The probability that a consumer uses a plastic card when making a purchase is .37.
- Given that the consumer uses a plastic card, there is a .19 probability that the consumer is 18 to 24 years old.
- Given that the consumer uses a plastic card, there is a .81 probability that the consumer is more than 24 years old.

U.S. Census Bureau data show that 14% of the consumer population is 18 to 24 years old.

- Given the consumer is 18 to 24 years old, what is the probability that the consumer use a plastic card?
 - Given the consumer is over 24 years old, what is the probability that the consumer uses a plastic card?
 - What is the interpretation of the probabilities shown in parts (a) and (b)?
 - Should companies such as Visa, MasterCard, and Discover make plastic cards available to the 18 to 24 year old age group before these consumers have had time to establish a credit history? If no, why? If yes, what restrictions might the companies place on this age group?
38. A Morgan Stanley Consumer Research Survey sampled men and women and asked each whether they preferred to drink plain bottled water or a sports drink such as Gatorade or Propel Fitness water (*The Atlanta Journal-Constitution*, December 28, 2005). Suppose 200 men and 200 women participated in the study, and 280 reported they preferred plain bottled water. Of the group preferring a sports drink, 80 were men and 40 were women.

Let

M = the event the consumer is a man

W = the event the consumer is a woman

B = the event the consumer preferred plain bottled water

S = the event the consumer preferred sports drink

- What is the probability a person in the study preferred plain bottled water?
- What is the probability a person in the study preferred a sports drink?
- What are the conditional probabilities $P(M | S)$ and $P(W | S)$?
- What are the joint probabilities $P(M \cap S)$ and $P(W \cap S)$?
- Given a consumer is a man, what is the probability he will prefer a sports drink?
- Given a consumer is a woman, what is the probability she will prefer a sports drink?
- Is preference for a sports drink independent of whether the consumer is a man or a woman? Explain using probability information.

